

(1) Murphy 1.12

(2) Murphy 1.34

(3) Murphy 2.11

(4) A chemical reaction $A \rightarrow B$ is carried out in a closed vessel. The following data are taken for the concentration of A, $C_A(g/L)$, as a function of time, $t(min)$, from the start of the reaction:

$t(min)$	0	36	65	100	160	∞
$C_A(g/L)$	0.1823	0.1453	0.1216	0.1025	0.0795	0.0495

A proposed reaction mechanism predicts that C_A and t should be related by the expression

$$\ln \frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = -kt$$

where k is a reaction rate constant, and the zero and infinity subscripts on the concentrations indicate the initial and very long time concentration values.

- (a) Do the data support this prediction? If so, determine the value of k .
- (b) If the tank volume is 30.5 gallons and there is no B in the tank at $t = 0$, how much B(g) does the tank contain in two hours?

(5) A mixture of ethanol and water contains 60 wt% water.

- (a) Assuming volume additivity of the components, estimate the specific gravity of the mixture at 20°C. What volume (in liters) of this mixture is required to provide 150 moles of ethanol?
- (b) Repeat part (a) with the additional information that the specific gravity of the mixture at 20°C is 0.93518 (making it unnecessary to assume volume additivity). What percentage error arises from the volume additivity assumption?